Hierarchical Waste Classification System

Audited Publication-Quality Evaluation Report

Generated: 2026-08-06 22:46:49 | GPU: NVIDIA GeForce RTX 3050 Laptop GPU | CUDA 12.8 | PyTorch 2.11.0+cu128

Hardware Environment

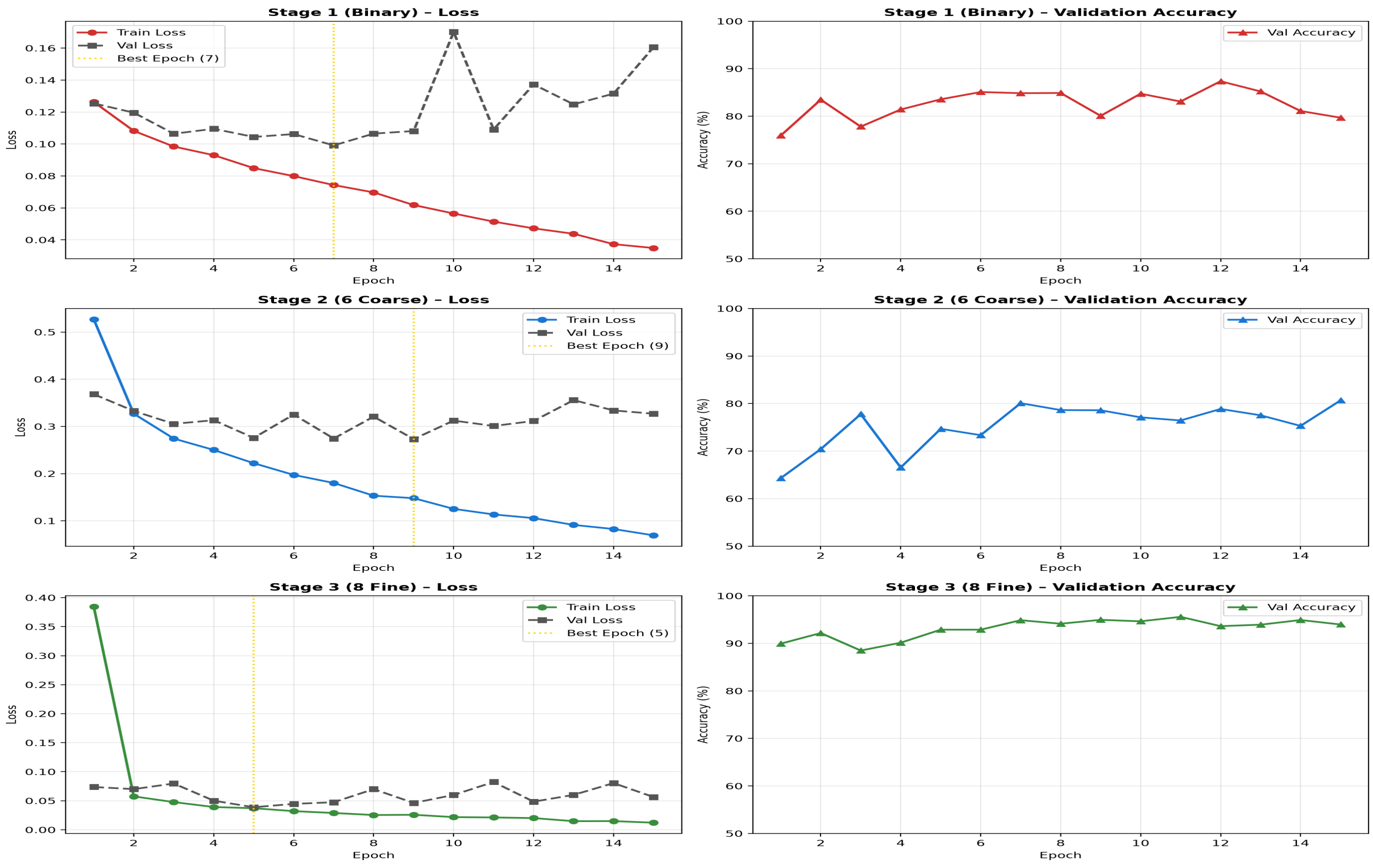
Property	Value
CPU	`AMD Ryzen 5 5600H with Radeon Graphics`
System RAM	19.86 GB
GPU	`NVIDIA GeForce RTX 3050 Laptop GPU`
GPU VRAM (total)	4.0 GB
CUDA Version	`12.8`
PyTorch Version	`2.11.0+cu128`
OS	`Windows-11-10.0.26200-SP0`

Training Statistics (from log timestamps)

Stage	Duration	Epochs	Best Ep (Val Loss)	Best Ep (Val Acc)	Final Train Loss	Final Val Loss	Final Val Acc
Stage 1	0:20:10	15	Ep 7 (0.0991)	Ep 12 (87.36%)	0.0349	0.1606	79.69%
Stage 2	0:20:08	15	Ep 9 (0.2732)	Ep 15 (80.67%)	0.0692	0.3270	80.67%
Stage 3	0:20:08	15	Ep 5 (0.0388)	Ep 11 (95.58%)	0.0122	0.0564	93.97%

Total training time (all 3 stages): 1:00:27

Training Curves (Loss & Accuracy)


Stage 1 (Binary) – Loss
Train Loss
Val Loss
Best Epoch (7)
Loss
Epoch
Stage 1 (Binary) – Validation Accuracy
Val Accuracy
Accuracy (%)
Epoch
Stage 2 (6 Coarse) – Loss
Train Loss
Val Loss
Best Epoch (9)
Loss
Epoch
Stage 2 (6 Coarse) – Validation Accuracy
Val Accuracy
Accuracy (%)
Epoch
Stage 3 (8 Fine) – Loss
Train Loss
Val Loss
Best Epoch (5)
Loss
Epoch
Stage 3 (8 Fine) – Validation Accuracy
Val Accuracy
Accuracy (%)
Epoch

Evaluation Metrics (Test Set — 2,568 Images)

All metrics computed from scratch from model checkpoints + test split.

Stage 1 (Binary)

Metric	Hierarchical CNN	KNN Baseline	∆ (abs pp or units)
Accuracy	79.5171%	74.9611%	+4.56 pp
Balanced Accuracy	81.6769%	50.0000%	+31.68 pp
MCC	0.5640	0.0000	+0.5640
Cohen's Kappa	0.5373	0.0000	+0.5373
Precision (Macro)	0.7511	0.3748	+37.63 pp
Recall (Macro)	0.8168	0.5000	+31.68 pp
F1-Score (Macro)	0.7638	0.4284	+33.53 pp
ROC-AUC	0.9033	0.5000	+0.4033
PR-AUC (Macro)	0.8686	—	—

Per-Class Metrics — Stage 1 (Binary)

Class	Precision	Recall	F1-Score	Support
biodegradable	0.5592	0.8600	0.6777	643
non_biodegradable	0.9430	0.7735	0.8499	1925

Stage 2 (6 Coarse)

Metric	Hierarchical CNN	KNN Baseline	∆ (abs pp or units)
Accuracy	66.9782%	44.1978%	+22.78 pp
Balanced Accuracy	62.1055%	16.6667%	+45.44 pp
MCC	0.5705	0.0000	+0.5705
Cohen's Kappa	0.5633	0.0000	+0.5633
Precision (Macro)	0.5882	0.0737	+51.45 pp
Recall (Macro)	0.6211	0.1667	+45.44 pp
F1-Score (Macro)	0.5882	0.1022	+48.60 pp
ROC-AUC	0.8344	0.5000	+0.3344
PR-AUC (Macro)	0.6065	—	—

Per-Class Metrics — Stage 2 (6 Coarse)

Class	Precision	Recall	F1-Score	Support
paper_cardboard	0.5297	0.7700	0.6276	487
organic	0.4555	0.8205	0.5858	156
glass	0.6630	0.5479	0.6000	334
metal	0.3962	0.4145	0.4051	152
plastic	0.5617	0.4342	0.4898	304
textile_battery	0.9230	0.7392	0.8209	1135

Stage 3 (8 Fine-grained)

Metric	Hierarchical CNN	KNN Baseline	∆ (abs pp or units)
Accuracy	63.3956%	3.0374%	+60.36 pp

Balanced Accuracy	59.3726%	12.5000%	+46.87 pp
MCC	0.5536	0.0000	+0.5536
Cohen's Kappa	0.5466	0.0000	+0.5466
Precision (Macro)	0.5492	0.0038	+54.54 pp
Recall (Macro)	0.5937	0.1250	+46.87 pp
F1-Score (Macro)	0.5551	0.0074	+54.78 pp
ROC-AUC	0.8251	0.5000	+0.3251
PR-AUC (Macro)	0.5320	—	—

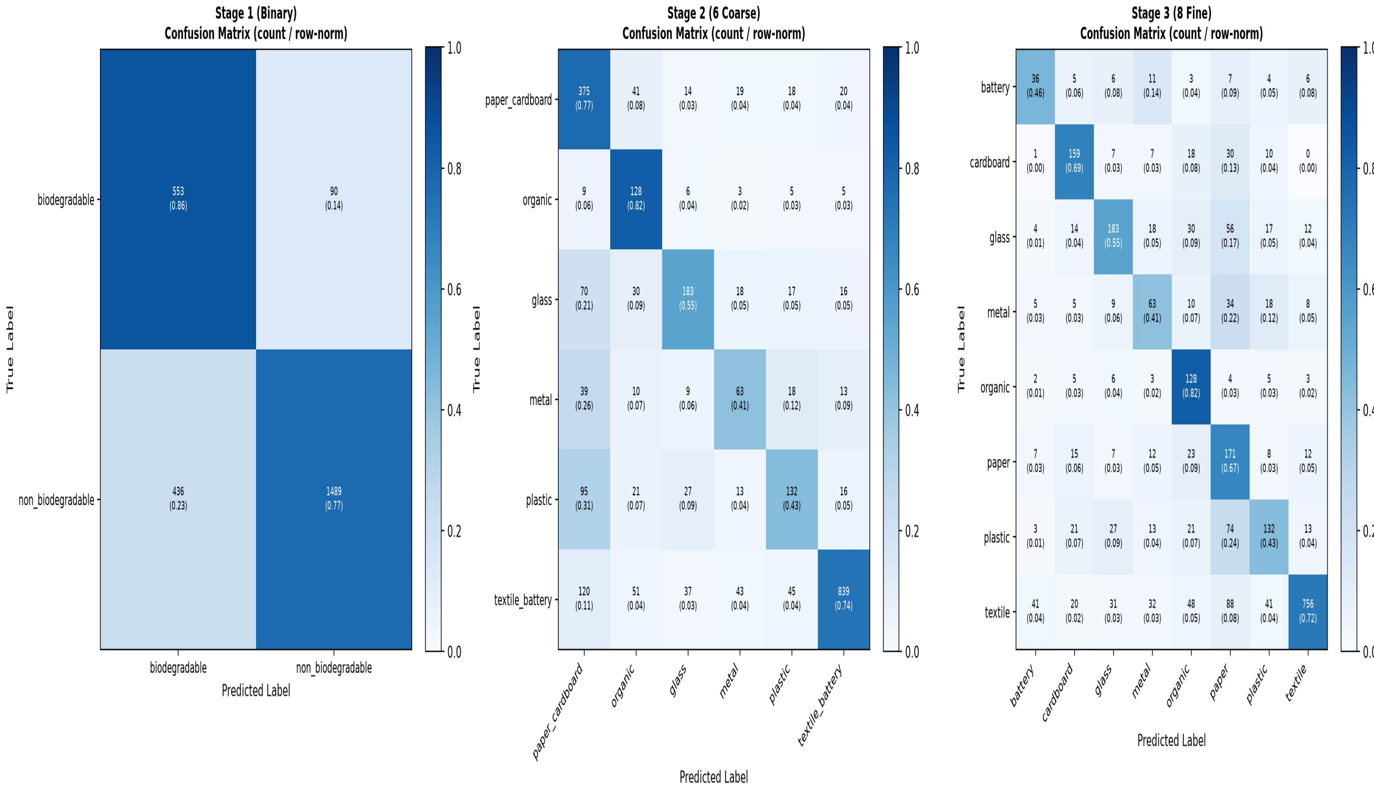
Per-Class Metrics — Stage 3 (8 Fine-grained)

Class	Precision	Recall	F1-Score	Support
battery	0.3636	0.4615	0.4068	78
cardboard	0.6516	0.6853	0.6681	232
glass	0.6630	0.5479	0.6000	334
metal	0.3962	0.4145	0.4051	152
organic	0.4555	0.8205	0.5858	156
paper	0.3685	0.6706	0.4757	255
plastic	0.5617	0.4342	0.4898	304
textile	0.9333	0.7152	0.8099	1057

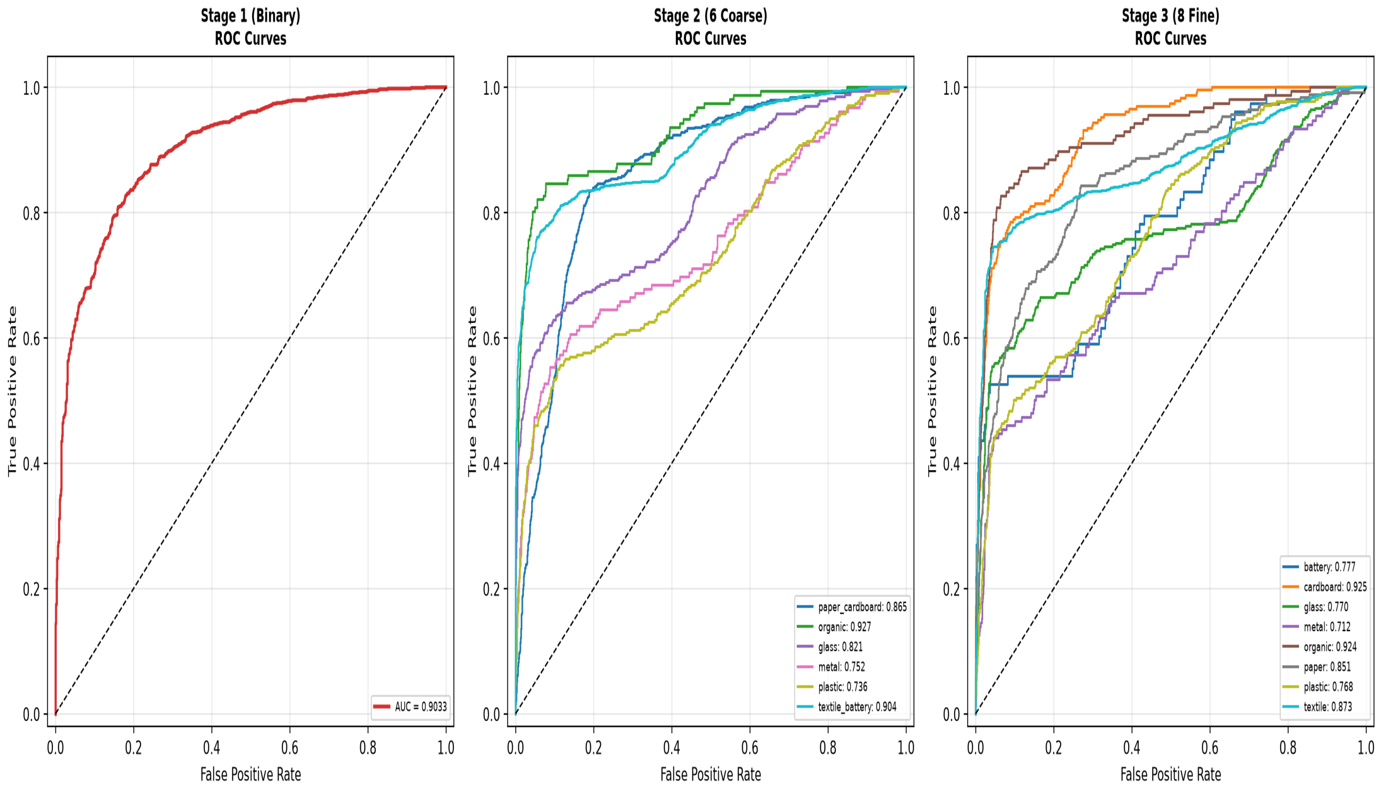
Inference Performance Benchmarks

Metric	Hierarchical CNN	KNN Baseline
Mean latency	8.367 ms	0.315 ms
p50 latency	8.093 ms	—
p95 latency	10.218 ms	—
p99 latency	11.670 ms	—
Throughput (FPS)	119.5	3179.4
Batch latency (batch=64, median)	428.55 ms	—
Batch throughput	149 img/sec	—
Peak VRAM (eval)	38.6 MB	N/A
Peak RAM (eval)	1193.2 MB	—
CPU utilization (1s window)	9.6%	—

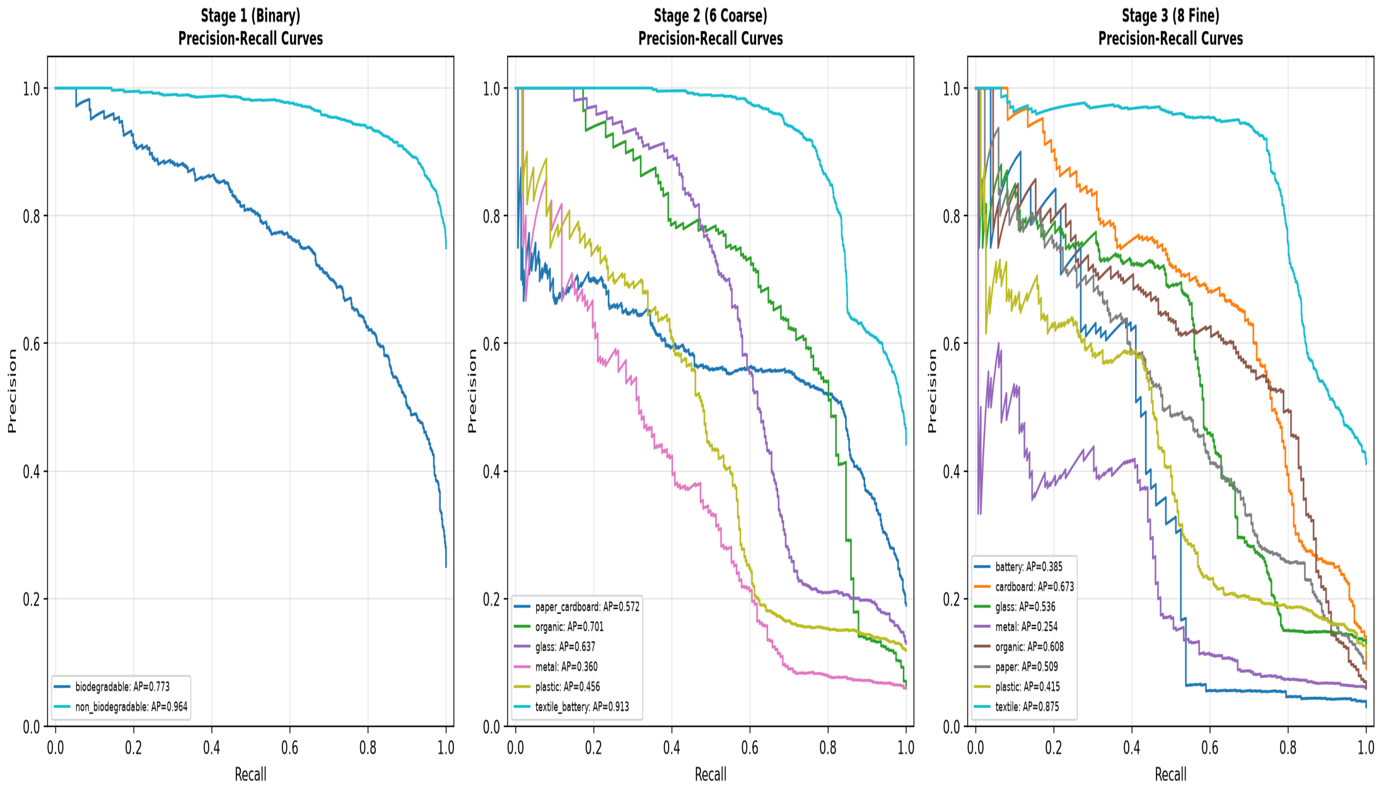
Confusion Matrices (count / row-normalized) — Stage 1, 2, 3



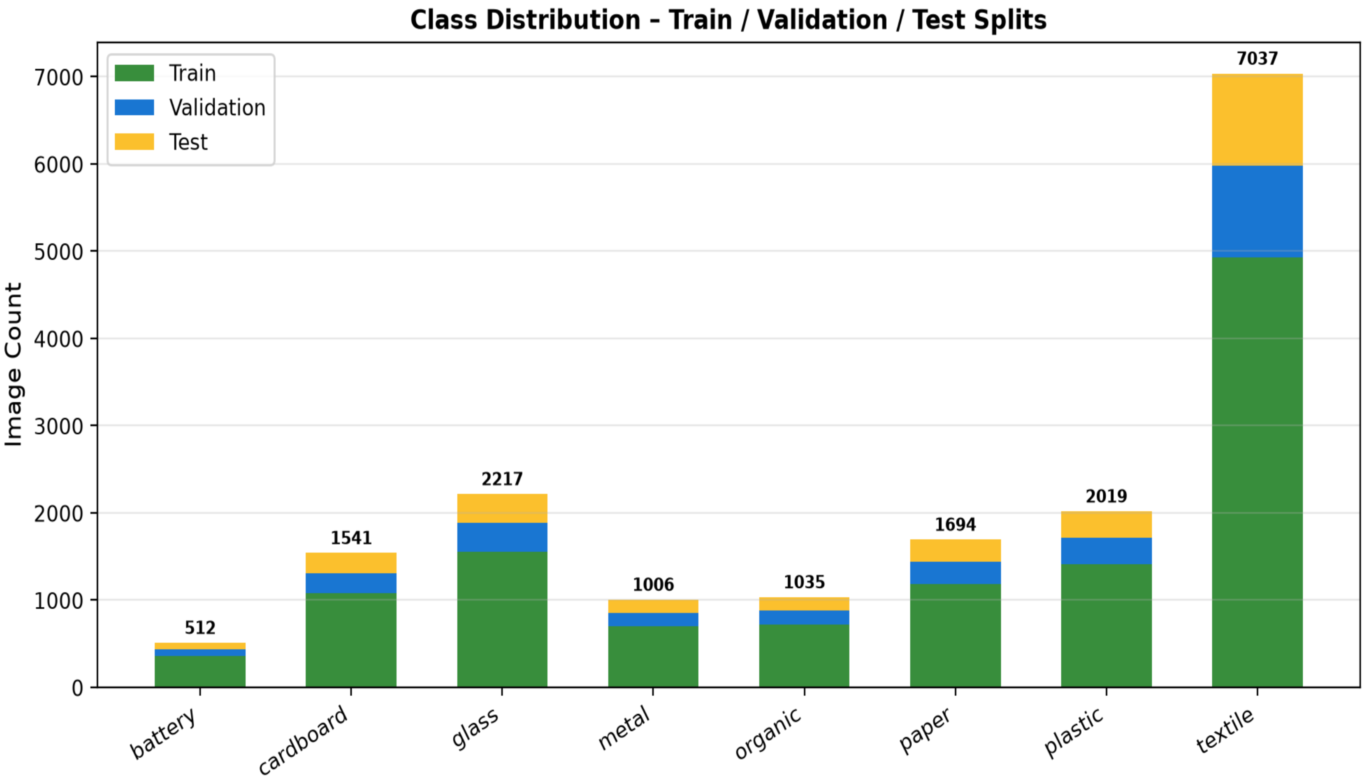
ROC Curves — Stage 1, 2, 3



Precision-Recall Curves — Stage 1, 2, 3



Dataset Class Distribution across Splits



Stage 3 Per-Class Test Accuracy